

{ "result": "The code you provided will currently result in a **ReferenceError** because the variables **a** and **b** are not defined inside the function or passed to it as arguments.\n\nTo make this function work correctly, you should pass **a** and **b** as **parameters**.\n\n### 1. The Standard Way (Recommended)\n\nYou define the variables inside the parentheses so the function can receive values when called.\n\njavascript\nfunction sum(a, b) {\n return a + b;\n}\n\n// How to use it:\nconsole.log(sum(5, 10)); // Output: 15\n\n### 2. The Modern Way (Arrow Function)\n\nIn modern JavaScript (ES6+), you can write this more concisely:\n\njavascript\nconst sum = (a, b) => a + b;\nconsole.log(sum(10, 20)); // Output: 30\n\n### 3. Using Default Parameters\n\nIf you want the function to work even if you forget to provide one of the numbers, you can set default values:\n\njavascript\nfunction sum(a = 0, b = 0) {\n return a + b;\n}\nconsole.log(sum(5)); // Output: 5 (5 + 0)\n\n### Why your original code might fail:\n\nIf you try to run `function sum(){ return a+b; }` without declaring **a** and **b** globally, the browser/environment will say: `Uncaught ReferenceError: a is not defined`. \n\nThe only way your original code works is if **a** and **b** are declared **outside** the function (Global Scope), like this:\n\njavascript\nlet a = 5;\nlet b = 10;\nfunction sum() {\n return a + b;\n}\n\n// This works, but is considered \"bad practice\"\n\nIt is better to pass variables directly into the function (as shown in Step 1) to keep your code clean and predictable." }